Skin Cancer Detection Using Deep Learning

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***Abstract*—This paper investigates skin cancer detection using deep learning. The final dataset configuration, model architecture, training protocol, and experimental results are pending confirmation and are therefore not claimed here.**

***Index Terms*—skin cancer, dermoscopy, deep learning, convolutional neural network, transfer learning**

I. Introduction

Automated analysis of skin-lesion images has been studied with conventional machine learning, convolutional neural networks (CNNs), transfer learning, and ensemble learning. The provided review sources describe the use of deep learning for skin-cancer detection and identify recurrent considerations including data availability, class imbalance, image quality, segmentation, and evaluation design [1], [2], [3], [4].

The provided empirical studies use dermoscopic-image datasets, including ISIC-based datasets and HAM10000, but their tasks and reported evaluation settings differ [5], [6], [7], [8], [9]. Accordingly, this work does not treat reported figures from prior studies as results of the proposed system.

This paper will document a reproducible experimental pipeline once the project-specific dataset split, architecture, and training configuration are supplied. Its results section intentionally contains placeholders rather than unsupported performance claims.

II. Related Work

A. Deep Learning and Transfer Learning

The source set contains studies that apply CNNs directly to lesion images [6], [7] and studies that compare or fine-tune pretrained networks [8], [9], [10]. Jain *et al.* compare six transfer-learning networks on HAM10000 and report image replication as their class-frequency intervention [8]. Gouda *et al.* evaluate a designed CNN alongside ResNet50, InceptionV3, and Inception-ResNet on their stated ISIC 2018 configuration [7]. These are distinct experimental settings and should not be compared as if they used an identical split.

B. Ensembles, Metadata, and Classical Pipelines

Imran *et al.* combine VGG, CapsNet, and ResNet learners [11]. Daghrir *et al.* study a hybrid of deep and classical machine-learning techniques for melanoma detection [12]. Ahmadi Mehr and Ameri incorporate lesion-image information with stated patient metadata using Inception-ResNet-v2 [13]. In contrast, Monika *et al.* use Dull Razor, filtering, segmentation, hand-crafted ABCD/GLCM features, and a multiclass SVM [14]. Together, these studies show that the source set spans end-to-end and manually engineered pipelines.

C. Research Gap

The provided reviews discuss class imbalance, dataset limitations, and concerns surrounding generalization or practical deployment [1], [2], [3], [4]. The empirical papers also differ in dataset composition, class definition, preprocessing, and metrics [6], [7], [8], [9], [10]. A source-supported gap for this project is therefore a transparent, project-specific comparison under one verified dataset split and reporting protocol. This is an investigation target, not a claim that a proposed model has solved the issue.

D. Recent Transformer and Imbalance-Aware Methods

The expanded source set includes vision-transformer and hybrid approaches for classification, segmentation, or both [15], [16], [17], [18], [19], [20]. It also includes studies that address imbalance through augmentation, focal loss, or synthetic-data generation [21], [22], [23], [24]. These methods use different datasets, target tasks, and evaluation procedures; their reported scores are retained in the source matrix rather than pooled into a single comparison.

Recent benchmarks and comparative studies evaluate transfer-learning architectures, multi-network ensembles, explainability, or hybrid feature pipelines [25], [26], [27], [28], [29], [30]. Their inclusion broadens the related-work coverage, but does not establish a common performance ranking without a shared experimental protocol.

III. Dataset and Data Preprocessing

A. Dataset

[DATASET NAME AND VERSION TO BE PROVIDED]
[DATASET SIZE, CLASS DEFINITIONS, LICENSE, AND PROJECT-SPECIFIC TRAIN/VALIDATION/TEST SPLIT TO BE VERIFIED]

B. Preprocessing

[IMAGE SIZE, NORMALIZATION, HAIR/ARTEFACT HANDLING, SEGMENTATION POLICY, AND AUGMENTATION POLICY TO BE PROVIDED]

The project configuration must not be inferred from the source papers' configurations.

IV. Proposed Methodology

A. Model Architecture

[MODEL ARCHITECTURE TO BE PROVIDED]

B. Training Procedure

[OPTIMIZER, LEARNING RATE, BATCH SIZE, EPOCHS, EARLY STOPPING, HARDWARE, AND RANDOM-SEED POLICY TO BE PROVIDED]

TABLE I
PROJECT RESULTS PLACEHOLDER

Metric	Result
Accuracy	[TO BE FILLED AFTER EXPERIMENT]
Precision	[TO BE FILLED AFTER EXPERIMENT]
Recall	[TO BE FILLED AFTER EXPERIMENT]
F1-score	[TO BE FILLED AFTER EXPERIMENT]
AUC	[TO BE FILLED AFTER EXPERIMENT]

V. EXPERIMENTAL SETUP

[EXPERIMENTAL PROTOCOL TO BE PROVIDED]

Evaluation will report only metrics computed on the project-specific held-out evaluation data. The target metrics are accuracy, precision, recall, F1-score, AUC, and a confusion matrix, where applicable.

VI. RESULTS AND DISCUSSION

No project-specific experimental result has been provided. Consequently, this section contains no numerical performance claim.

VII. CONCLUSION

This source-grounded draft separates findings reported in prior literature from the project methodology and results. Dataset, implementation, and experiment details are required before a final methodology or results claim can be made.

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